

- 5 (a) An object travels in a circle at constant speed.

State the names of **two** quantities that vary during the motion of the object.

1

2 [2]

- (b) A charged particle of mass m and with charge q enters a region of uniform magnetic field, perpendicular to the field lines. The magnetic flux density is B .

The particle travels in a circle with period T and radius r .

- (i) By considering the magnetic force acting on the particle, show that

$$B = \frac{2\pi m}{qT}.$$

[3]

- (ii) The particle is an alpha particle. The period of the circular motion is $2.5\mu\text{s}$.

Calculate B .

$$B = \dots\dots\dots \text{T} \quad [2]$$

- (iii) A second alpha particle is in the same uniform field. It travels in a circle of radius $2r$.

State and explain how the periods of the motion of the two particles compare.

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..... [1]

- (iv) The speed of the alpha particle in (b)(ii) is $1.1 \times 10^6 \text{ m s}^{-1}$. An electric field is applied so that this particle now moves with constant velocity.

Use your answer in (b)(ii) to calculate the electric field strength E . Give the unit with your answer.

$E =$ unit [2]

[Total: 10]